National Taiwan University

Geodynamics

Assignment 1

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Question 1

(a) Using information provided in Slide #31 and #32 of Week 1 Lecture, compute the heat production rate (H and P) of tholeiitic basalt (i.e. normal mid ocean ridge basalt).

$$H_{\text{BSE}} = \sum_E H_E^* C_E \tag{1}$$

And from the table I have:

$$\mathsf{U} = 0.07\,\text{ppm}, \quad \mathsf{Th} = 0.19\,\text{ppm}, \quad \mathsf{K} = 0.088\% \tag{2}$$

and the radiation heat is:

$$\mathsf{U}\ H^* = 9.81 \times 10^{-5}\,\text{W}\,\text{kg}^{-1} \tag{3}$$
$$^{232}\mathsf{Th}\ H^* = 2.64 \times 10^{-5}\,\text{W}\,\text{kg}^{-1} \tag{4}$$
$$\mathsf{K}\ H^* = 3.48 \times 10^{-9}\,\text{W}\,\text{kg}^{-1} \tag{5}$$

So the resultant radiation heat is

$$H = H_U^* C_U + H_{Th}^* C_{Th} + H_K^* C_K \tag{6}$$
$$= 0.07 \cdot 10^{-6} \times 9.81 \cdot 10^{-5} + 0.19 \cdot 10^{-6} \times 2.64 \cdot 10^{-5} + 0.088 \cdot 10^{-2} \times 3.48 \cdot 10^{-9} \tag{7}$$
$$= 1.5 \times 10^{-11}\,\text{W}\,\text{kg}^{-1} \tag{8}$$
$$P = \rho \times H = 2900\,\text{kg}\,\text{m}^{-3} \cdot 1.5 \times 10^{-11} = 4.3 \times 10^{-8}\,\text{W}\,\text{m}^{-3} \tag{9}$$

(b) Compare the heat production rate H of granite, basalt, depleted peridotite and undepleted mantle. Are they consistent with the differentiation history of silicate Earth?

Table 1: Radiogenic Heat of the Rocks

Rock Type	ρ (kg m^{-3})	H (W kg^{-1})	P (W m^{-3})
Reference undepleted	3200	7.4×10^{-12}	2.4×10^{-8}
Peridotite (depleted)	3200	3.1×10^{-13}	9.9×10^{-10}
Basalt (tholeiitic)	2900	1.5×10^{-11}	4.3×10^{-8}
Average continental crust	2700	3.4×10^{-10}	9.1×10^{-7}

From Table 1, we obtain

$$\mathsf{H}_{\text{continental crust}} \gg \mathsf{H}_{\text{basalt}} > \mathsf{H}_{\text{undepleted mantle}} \gg \mathsf{H}_{\text{depleted peridotite}}$$

This ordering is consistent with the differentiation history of the silicate Earth. The main heat-producing elements (U, Th, and K) are *incompatible elements*, meaning they preferentially partition into the *melt* during partial melting rather than remaining in mantle minerals.

Silicate-Earth differentiation transfers incompatible, heat-producing elements from the mantle into melts and ultimately the crust, leaving the residual mantle depleted, which matches the trend observed in Table 1.

Question 2

(a) Computing the temperature profile of a continental region. Assuming the following parameters: steady state, thermal conductivity k continental heat production rate $P = \rho H_0 e^{-\frac{z}{d}}$, d is much smaller than the thickness of the continent, the mantle heat flux q_m and the surface temperature T_0.

$$T = -\frac{1}{k} \iint P(z)\,\mathrm{d}z^2 \tag{10}$$

First, I have

$$q(\infty) = -q_m \tag{11}$$

then

$$q(\infty) - q(z) = \int_z^\infty P(\xi)\,\mathrm{d}\xi \Rightarrow q(z) = -q_m - \int_z^\infty \rho H_0 e^{-\frac{\xi}{d}}\,\mathrm{d}\xi \tag{12}$$

$$\Rightarrow q(z) = -q_m - \rho H_0 d\, e^{-\frac{z}{d}} \tag{13}$$

And the second integral η is just a integral variable.

$$T(z) - T_0 = \int_0^z \frac{\mathrm{d}T}{\mathrm{d}\eta}\,\mathrm{d}\eta = -\frac{1}{k}\int_0^z \left(-q_m - \rho H_0 \mathrm{d}\, e^{-\frac{\eta}{d}}\right)\,\mathrm{d}\eta \tag{14}$$

$$\Rightarrow T(z) = T_0 + \frac{q_m}{k} z + \frac{\rho H_0 d^2}{k}\left(1 - e^{-\frac{z}{d}}\right) \tag{15}$$

(b) Plot the temperature profile from surface to 300 km depth by setting $k = 3\,\mathrm{W\,m^{-1}\,K^{-1}}$, $d = 7\,\mathrm{km}$, $\rho H_0 = 2 \times 10^{-6}\,\mathrm{W\,m^{-3}}$, $T_0 = 0\,°\mathrm{C}$, $q_m = 0.04\,\mathrm{W\,m^{-2}}$

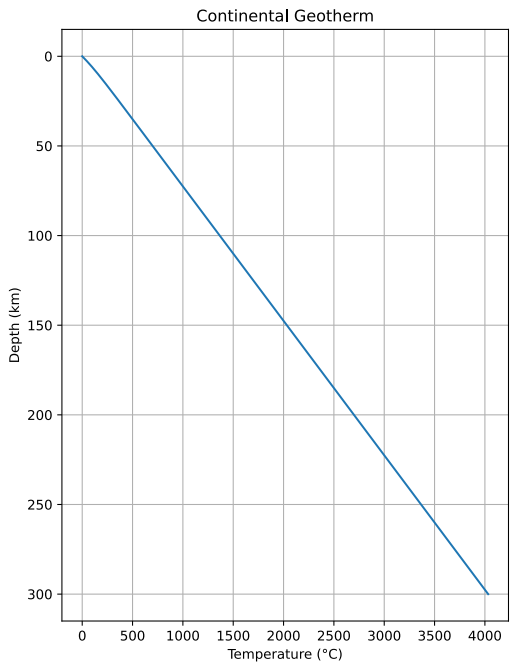

Figure 1: Temperature profile from surface to 300 km.

Question 3

(a) Verify the solution of the half-space cooling problem in Slide #38 of Week 1 Lecture by substituting the solution into the thermal diffusion equation.

$$T(t,z) = T_0 + (T_m - T_0)\,\mathrm{erf}\left(\frac{z}{2\sqrt{\kappa t}}\right) \tag{16}$$

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial z^2} \tag{17}$$

$$\frac{\partial T}{\partial t} = (T_m - T_0)\cdot \frac{\partial}{\partial t}\left[\mathrm{erf}\left(\frac{z}{2\sqrt{\kappa t}}\right)\right] \tag{18}$$

$$= (T_m - T_0)\cdot -\frac{ze^{-\frac{z^2}{4\kappa t}}}{2\sqrt{\pi\kappa t^3}} \tag{19}$$

$$\frac{\partial^2 T}{\partial z^2} = (T_m - T_0)\cdot \frac{\partial}{\partial z}\left(\frac{\partial}{\partial z}\left[\mathrm{erf}\left(\frac{z}{2\sqrt{\kappa t}}\right)\right]\right) \tag{20}$$

$$= (T_m - T_0)\cdot \frac{\partial}{\partial z}\left(\frac{e^{-\frac{z^2}{4\kappa t}}}{\sqrt{\pi\kappa t}}t\right) \tag{21}$$

$$= (T_m - T_0)\cdot -\frac{ze^{-\frac{z^2}{4\kappa t}}}{2\sqrt{\pi}\kappa^{\frac{3}{2}}t^{\frac{3}{2}}} \tag{22}$$

$$\Rightarrow \kappa\cdot\frac{\partial^2 T}{\partial z^2} = (T_m - T_0)\cdot -\frac{ze^{-\frac{z^2}{4\kappa t}}}{2\sqrt{\pi}\kappa^{\frac{1}{2}}t^{\frac{3}{2}}} \tag{23}$$

(b) When solving the thermal profile of oceanic plate in Slide #36, we assumed that the thermal diffusion in the horizontal direction is negligible. Please verify this assumption by comparing the magnitude of $\frac{\partial^2 T}{\partial x^2}$ and $\frac{\partial^2 T}{\partial z^2}$. Hint: $x = vt$ and $v \approx 3\mathrm{cm/yr}$.

$$\frac{\partial^2 T}{\partial z^2} = (T_m - T_0)\cdot -\frac{ze^{-\frac{z^2}{4\kappa t}}}{2\sqrt{\pi}\kappa^{\frac{3}{2}}t^{\frac{3}{2}}} \tag{24}$$

$$\frac{\partial^2 T}{\partial x^2} = (T_m - T_0)\cdot \frac{\partial}{\partial x}\left(\frac{\partial}{\partial x}\left[\mathrm{erf}\left(\frac{z}{2\sqrt{\frac{x\kappa}{v}}}\right)\right]\right) \tag{25}$$

$$= (T_m - T_0)\cdot \frac{\sqrt{v}z\left(6\kappa x - vz^2\right)e^{-\frac{vz^2}{4\kappa x}}}{8\sqrt{\pi}\kappa^{\frac{3}{2}}x^{\frac{7}{2}}} \tag{26}$$

Verify the assumption by calculating[1]

$$\left|\frac{\frac{\partial^2 T}{\partial x^2}}{\frac{\partial^2 T}{\partial z^2}}\right| = \frac{t^{\frac{3}{2}}\sqrt{v}\left(-6\kappa x + vz^2\right)e^{\frac{z^2\left(-\frac{v}{x}+\frac{1}{t}\right)}{4\kappa}}}{4x^{\frac{7}{2}}} \tag{27}$$

Since the oceanic plate moves with constant velocity $v, x = vt$. Then

$$\frac{v}{x} = \frac{v}{vt} = \frac{1}{t} \tag{28}$$

[1]This is calculated with SymPy. See https://www.sympy.org/en/index.html

Therefore the exponential term becomes

$$e^{\frac{z^2\left(-\frac{v}{x}+\frac{1}{t}\right)}{4\kappa}} = e^0 = 1 \tag{29}$$

So the ratio simplifies to

$$\left|\frac{\frac{\partial^2 T}{\partial x^2}}{\frac{\partial^2 T}{\partial z^2}}\right| = \left|\frac{t^{3/2}\sqrt{v}(-6\kappa x + vz^2)}{4x^{7/2}}\right| = \left|\frac{v^{3/2}t^{3/2}(-6\kappa t + z^2)}{4v^{7/2}t^{7/2}}\right| \tag{30}$$

$$= \left|\frac{z^2 - 6\kappa t}{4v^2t^2}\right| \tag{31}$$

For oceanic lithosphere:

$$\kappa \sim 10^{-6}\,\mathrm{m}^2/\mathrm{s}$$

$$v \sim 3\ \mathrm{cm/yr} \approx 10^{-9}\ \mathrm{m/s}$$

Take plate age:

$$t \sim 10^7\ \mathrm{yr}$$

$$\approx 3 \times 10^{14}\ \mathrm{s}$$

Thermal diffusion depth:

$$z \sim \sqrt{\kappa t}$$

$$\sim 10^4\ \mathrm{m}$$

Then

$$z^2 \sim 10^8$$

$$\kappa t \sim 10^8$$

No numerator:

$$z^2 - 6\kappa t \sim 10^8 - 6 \times 10^8 \sim 10^{7.5} \tag{32}$$

Denominator:

$$4v^2t^2 = 4(10^{-9})^2(10^{14})^2 \sim 10^{10} \tag{33}$$

So

$$\left|\frac{\partial^2 T/\partial x^2}{\partial^2 T/\partial z^2}\right| \sim \frac{10^{7.5}}{10^{10}} = 10^{-2.5} \ll 1 \tag{34}$$

$$\Rightarrow \left|\frac{\partial^2 T}{\partial x^2}\right| \ll \left|\frac{\partial^2 T}{\partial z^2}\right| \tag{35}$$

We are just about 0.3% off.

Question 4

Which type of melting process (decompression, excess heat, flux) produces the following volcanoes?

(a) Tatun volcano in Taipei

- Excess heat

(b) Iceland

- Decompression melting
- Excess heat

(c) Easter Island

- Excess heat

(d) Mt. Fuji

- Flux melting

(e) Mt. Kilimanjaro

- Decompression melting

(f) Galapago Islands

- Excess heat